

An-Chi Liu

SENIOR SOFTWARE ENGINEER · AUTONOMOUS DRIVING SIMULATION & INFRASTRUCTURE

Taiwanese, based in Tokyo, Japan (Permanent Resident)

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Work Experience

T2, Inc. (L4 Truck Autonomous Driving)

Senior Software Engineer

- Conceived, proposed, and built a company-wide **simulation QA platform** from scratch. Secured resources and drove adoption through user discovery and feedback.
- Architected a hybrid **Kubernetes** system across on-premises **GPU** and cloud workers. Automated **Docker** deployment and built a **Go** backend and **React** frontend.
- Integrated the platform with T2 services and **CI** workflows to orchestrate hundreds of daily simulation jobs across the autonomous-driving stack.
- Built a **validation framework** with **time-series signal processing** for simulation results across millions of scenarios. Eliminated thousands of engineer-hours of manual validation.
- Worked on planning system's **close-loop simulation** with **OpenDrive**-compatible mapping system and an **OpenScenario**-compatible scenario engine.

Hybrid
Nov. 2025 – Present

Datra AI

Technical Co-founder

- Co-founded an AI startup for **Shopify** merchants. Led system architecture, a **Go** backend, a **React** frontend, and deployment on **AWS** with **Docker**. Managed two developers.
- Built an AI system that integrated OpenAI and Claude models with prompt engineering and retrieval-augmented generation (**RAG**).

Remote
Feb. 2025 – Sep. 2025

Mujin, Inc.

Backend Software Engineer

- Developed a **C++ network device discovery service** with PcapPlusPlus. It finds **Ethernet/IP** robots, industrial cameras, and other edge devices within seconds.
- Built a live-streaming and playback system for **100+ surveillance cameras** with under **50 ms** end-to-end latency, using **FFmpeg**, **WebRTC**, and **Go**.
- Extended production **C++/Python** robot and warehouse control systems across real-time motion, sensor, and manipulation pipelines. Added Go web APIs, C++ ZMQ APIs, and Qt/QML operator interfaces.

Tokyo, Japan
Sep. 2022 – May 2025

Education

National Yang Ming Chiao Tung University (NYCU)

M.S. in Computer Science and Engineering

- GPA: 4.11/4.3. Thesis: **Offworker: An Offloading Framework for Parallel Web Applications**, published in **WISE 2022** (top-tier conference).
- Received the **Best Master's Thesis Award** from the Institute of Information & Computing Machinery, Taiwan (2022; one of 13 awardees).

Hsinchu, Taiwan
Sep. 2019 – May. 2022

National Taiwan University (NTU)

B.S. in Bio-Industrial Mechatronics Engineering

Taipei, Taiwan
Sep. 2014 – Jun. 2019

Open-Source Projects

solvcon

- Major contributor** to a 73-star **C++/Python** toolkit for PDE simulation, numerical computing, visualization, and AI-assisted engineering. [View GitHub PRs]
- Added **C++ time-series kernels** and **DataFrame** operators, plus native **MCAP** ingestion and schema-driven **ROS 2 CDR** decoding.
- Built an interactive **2D graphics** system for lines, Bezier curves, ellipses, and polygons to support numerical-simulation visualization.
- Developed an **AI agent** system integrated with the Pilot GUI. Strengthened Windows builds and CI, and presented the array work at **PyCon JP 2024**.

PcapPlusPlus

- One of five core maintainers** since 2024 for a 3.1k-star, multi-platform **C++** packet-processing library.
- Improved performance, modernized the **CI/CD** pipeline, added protocol support, and expanded documentation.

Skills

Programming

Proficient: C++11/14/17, Python 3; **Experience:** Go, Bash, JavaScript, TypeScript

Autonomy/Simulation

Closed-loop simulation, OpenSCENARIO, OpenDRIVE, scenario/benchmark design, planning-stack evaluation, evaluation metrics and dashboards, regression testing, log replay, motion and manipulation pipelines

Infrastructure/Network

Linux, on-premises GPU clusters, Kubernetes, Docker, AWS, Ethernet/IP, network device discovery, packet capture, PcapPlusPlus, ZMQ, WebRTC, CI/CD, CMake, GDB, Git

Data/AI Languages

MCAP, ROS 2, CDR, pybind11, NumPy-style arrays, AI-agent backends, OpenAI/Claude integration, RAG
Mandarin Chinese (Native), English (Business), Japanese (Limited Conversational)